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Data assimilation in a coupled physical-biological model for the Bohai Sea and the Northern Yellow Sea

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Abstract. One difficulty with coupled physical-biological ocean models is determining optimal values of poorly known model parameters. The variational adjoint assimilation method is a powerful tool for the automatic estimation of parameters. We used this method to incorporate remote-sensed chlorophyll-*a* data into a coupled physical-biological model developed for the Bohai Sea and the Northern Yellow Sea. A 3-D NPZD model of nutrients (N), phytoplankton (P), zooplankton (Z) and detritus (D) was coupled with a physical model, the Princeton Ocean Model. Sensitivity analysis was carried out to choose suitable control variables from the model parameters. Numerical twin experiments were then conducted to demonstrate whether the spatio-temporal resolutions of the observations were adequate for estimating values of the control variables. Finally, based on the success of the twin experiments, we included remote-sensed chlorophyll-*a* data in the NPZD model. With the adjoint assimilation of these chlorophyll-*a* data, the coupled model better describes spring and autumn phytoplankton blooms and the annual cycle of phytoplankton at the surface layer for the study area. The annual cycle of simulated surface nutrient concentrations also agreed well with field observations. The adjoint method greatly improves the modelling capability of coupled ocean models, helping us to better understand and model marine ecosystems.

Additional keywords: adjoint method, control variable, ecosystem.

Introduction

The coupled physical-biological models with various levels have recently been developed for different regions of the ocean (Franks and Chen 1996; Eslinger *et al.* 2000; Lenhart and Pätzsch 2001; Franks 2002; Baird *et al.* 2007). The main difficulty going with these models is to obtain an optimal estimate of the poorly measured or poorly known model parameters, which leads to a better agreement between model outputs and observations. Typically these parameters are individually and unsystematically adjusted until the model results agree with the observations. Such a subjective adjustment is an arduous task, even with an ecosystem model that has few parameters (Spitz *et al.* 1998). However, data assimilation techniques such as the simulated annealing (Matear 1995) or variational adjoint method (Lawson *et al.* 1995; Spitz *et al.* 1998; Friedrichs 2001a; Schartau *et al.* 2001) have been proven to be vital tools to adjust model parameters systematically and objectively, which develop quantitatively realistic ecosystem models with better accuracy, efficiency and prognostic ability. These techniques fix unknown parameters and save our time in searching parameter sets that lead to a better agreement between model results and observations. In addition, models' structures can also be improved based on the assimilation results.

However, there are very few studies on applications of data assimilation techniques to marine ecosystem models in China (Zhao 2002; Xu *et al.* 2005). In the present study, the variational

adjoint assimilation of chlorophyll-*a* data in a coupled physical-biological model for the Bohai Sea and the Northern Yellow Sea is expatiated, and two groups of data assimilation experiments were carried out. First, numerical twin experiments were run to determine if the model-generated data (synthetic observations), which have the same spatio-temporal resolutions as the remote sensed chlorophyll-*a* data are sufficient to estimate model parameters. The results of twin experiments were also used to determine if all the model parameters can be recovered and to test the capability of the adjoint method to recover the parameters. Then, based on the twin experiments, the variational adjoint assimilation of remote sensed chlorophyll-*a* data was conducted. Some preliminary results of the assimilation were analysed.

Materials and methods

Study area

The study area is the Bohai Sea (BS) and the Northern Yellow Sea (NYS) (34.17°N–41°N, 117.5°E–126.67°E). Figure 1 shows the bathymetric chart. The average water depth of the BS is ~18 m and that of the NYS is ~38 m. With adequate underwater illumination, high chlorophyll concentration and rich kinds of algae and plankton, the BS and NYS have unique resources and geographical locations that make them the important areas of aquaculture. However, with economic development, environmental pollution is increasing, which deteriorates the ecosystem

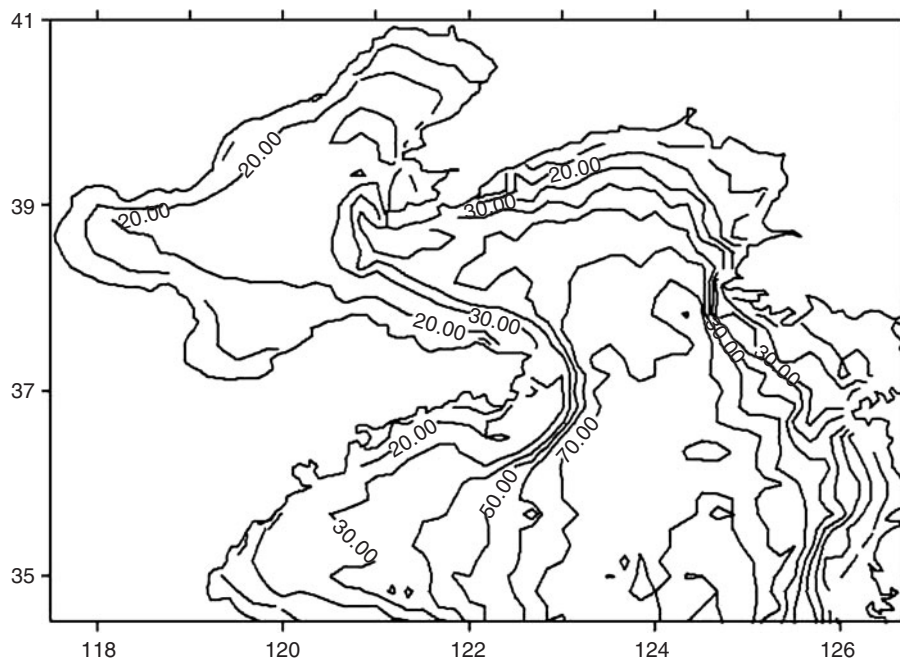


Fig. 1. Bathymetric chart of the Bohai Sea (BS) and the Northern Yellow Sea (NYS). (The *x*-axis denotes the east longitude, and *y*-axis denotes the north latitude. The isobaths is in unit m.)

of the study area. Thus it is of great significance to study the marine ecosystem using numerical models in order to protect the marine life and fisheries and improve the capability of making continuing use of the study area.

Coupled physical-biological model

A 3-D NPZD (nutrient-phytoplankton-zooplankton-detritus) biological model is coupled with a physical model for the study area.

The physical model is Princeton Ocean Model (POM). It is run with six vertical levels in a sigma-coordinate (Mellor 2004). The horizontal resolution is $1/6^\circ \times 1/6^\circ$ (longitude $\times$ latitude); the open boundary is along the most southern latitude. The numerical model is forced through the sea surface by heat flux and wind stress computed based on National Centers for Environmental Prediction/National Center for Atmospheric Research (NCEP/NCAR) reanalysis (<http://dss.ucar.edu/datasets/ds090.2/>). The initial conditions for salinity and temperature are the climatology data available from the National Oceanographic Data Center (NODC) Levitus – World Ocean Atlas 2001. The numerical model was forced along the open boundary by the surface tides (i.e. the elevation of M_2 , S_2 , K_1 and O_1 constituents) and circulation (i.e. current velocity from Ocean Circulation and Climate Advance Modelling (OCCAM) data) (Zhao *et al.* 2005). The circulation in the basin starts at rest.

The NPZD biological model is developed based on the NPZ (nutrient-phytoplankton-zooplankton) model of Franks and Chen (2001). Referring to the study of Zhao (2002) for the BS, we add a state variable named detritus. Thus a modified 3-D NPZD model is established for the BS and NYS. The four state variables are expressed in the unit of mmol N m^{-3} .

Figure 2 shows a schematic diagram of the biological interactions among the four state variables. The phytoplankton converts

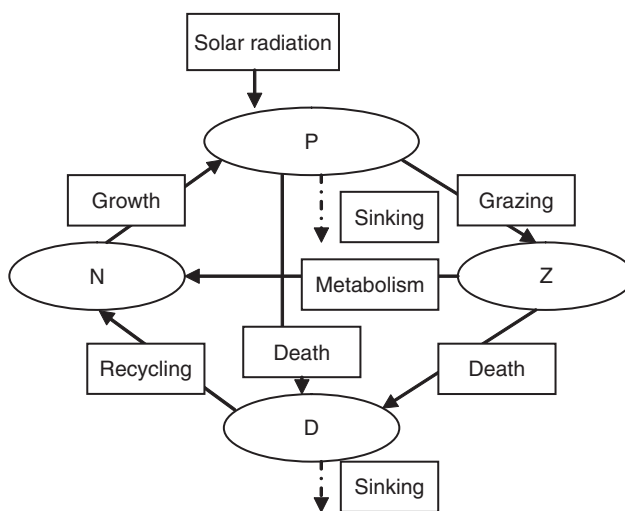


Fig. 2. Schematic diagram of the NPZD (nutrient-phytoplankton-zooplankton-detritus) model.

the absorbed solar energy and the nutrient to its self-material; the dead phytoplankton becomes a part of the detritus; the zooplankton assimilates the grazed phytoplankton into its self-material; the zooplankton releases the nutrient through the metabolism process; the dead zooplankton also becomes a part of the detritus; the nutrient is supplied by the mineralisation and recycle of the detritus.

The biological model is run in the year 2003, using initial conditions obtained by running the model until steady-state. The time step for the biological model is 30 min. The model is run in Cartesian coordinates, and there are six layers in the vertical, with layer depth of $10 \text{ m} \times 3$, $20 \text{ m} \times 1$, and $25 \text{ m} \times 2$

Table 1. Parameters in the NPZD (nutrient-phytoplankton-zooplankton-detritus) model

Parameter	Symbol	Initial value	Unit	Reference
Phytoplankton (P)				
Maximum uptake rate of nutrient by P	V_m	1.0	d^{-1}	Yoshimori <i>et al.</i> (1995)
Half-saturation constant of nitrogen by P	k_s	1.0	$(\text{mmol N m}^{-3})^{-1}$	Franks <i>et al.</i> (1986), Gao (1998), Zhao (2002)
Light extinction coefficient	k_{ext}	0.1	m^{-1}	Zhao (2002)
Sinking rate of P	w_P	3.4	d^{-1}	Zhao (2002)
Mortality rate of P	d_P	0.1	d^{-1}	Franks and Chen (1996), Yoshimori <i>et al.</i> (1995)
Temperature adjustment coefficient for growth of P	AQ_{10}	2.08	/	Fransz and Verhagen (1985)
Zooplankton (Z)				
Ivlev grazing constant	λ	0.2	$(\text{mmol N m}^{-3})^{-1}$	Franks and Chen (1996), Zhao (2002)
Maximum grazing rate of P by Z	G_m	0.5	d^{-1}	Franks and Chen (1996), Zhao (2002)
Assimilation coefficient of Z	γ	0.75	/	Gunson <i>et al.</i> (1999), Zhao (2002)
Excreting rate of Z	θ	0.03	/	Gunson <i>et al.</i> (1999)
Mortality rate of Z	d_Z	0.2	d^{-1}	Franks and Chen (1996), Gunson <i>et al.</i> (1999)
Temperature adjustment coefficient for growth of Z	BQ_{10}	3.1	/	Fransz and Verhagen (1985)
Detritus (D)				
Regeneration rate of D	e	0.05	d^{-1}	Gunson <i>et al.</i> (1999), McCreary <i>et al.</i> (1996)
Sinking rate of D	w_D	5.0	d^{-1}	Gunson <i>et al.</i> (1999)

respectively. The water velocities, water temperature, and the eddy diffusivities calculated by POM are converted from sigma-coordinates to Cartesian coordinates and provided for the NPZD model. The NPZD model includes local inputs of the nutrient by river discharge (the Yellow River, Luan River and Yalujiang River) and atmospheric sedimentation within the domain of the study.

The NPZD model equations are listed in Appendix A. There are 14 model parameters in the NPZD model, whose values are poorly measured or poorly known. Although the values of the parameters are not precisely known, many published reports can provide reasonable initial estimates of these values and associated uncertainties. Therefore, we prescribe the initial values of the parameters based on various investigations, in which their conceptual model is similar to our NPZD model, or the domain area is within the same latitudes as the study area, i.e. the ocean environments are similar. Table 1 lists the initial values and units of the parameters and the references. The initial values are not completely the same as those in the references. Values are adjusted within the ranges given by different investigators in order to reproduce qualitatively through simulation the phytoplankton blooms in the spring and autumn and annual cycle of nutrients for the studied domain (see Figs 8 and 10).

Variational adjoint method

In the present study, the variational adjoint method (adjoint method for simplicity) is applied to estimate the optimal values

of the model parameters (Bennet 1992; Wunsch 1996), which has been recently widely used in marine ecosystem models. Typically, the adjoint model can be established either by the adjoint operator method or by the Lagrange method, which was used in the present study (see Appendix B for the adjoint equations of the NPZD model. With the Lagrange method, the adjoint equations were derived from Eqns (A-1) to (A-4) (Lv and Liu 2001)).

The NPZD model is run forward in time using the first-guess parameter values (those listed in Table 1), in order to obtain a value of the cost function, which is defined to be a measure of the misfit between the available state variable observations ($\hat{C}$) and the corresponding model simulations (C) run with the set of parameter values X . If the errors in the data to be assimilated are uncorrelated, the cost function is formulated as a weighted sum of error squares (Lawson *et al.* 1996; Friedrichs 2001a; Garcia-Gorriz *et al.* 2003):

$$J = \frac{1}{2} \sum_{i=1}^M \sum_{j=1}^N W_{ij} [C_{ij}(X) - \hat{C}_{ij}]^2, \quad (1)$$

where M is the number of state variables for which the observations are available; N is the model time-steps; and the subscripts i and j denote the horizontal coordinates of the grid point. The weight, W , accounts for the differences in the relative magnitudes of the time averages of each observed state variable. Similar to the case in Garcia-Gorriz *et al.* (2003), only chlorophyll-*a*

concentrations are assimilated in the present study, therefore $M = 1$, and we assume that the weight has the same value, $W = 1$.

An adjoint or backward model is used to compute the gradients of the cost function with respect to the input parameters, frequently referred to as the 'control variables'. Values of these gradients are used in an optimisation procedure (the steepest descent algorithm in the present study) to compute the optimal direction towards the minimum of the cost function, and the optimal size in that direction (Friedrichs 2001a). The model is then rerun with the new values of the control variables that yield a smaller misfit between the model outputs and observations. This process is carried out in an iterative manner for either a specific number of iterations, or when a very small value of the cost function is obtained or a specified convergence criterion is satisfied (Spitz *et al.* 1998). Those are the optimal values of the control variables recovered by the adjoint method.

Numerical twin experiments

A numerical twin experiment with simulated synthetic data is conducted to test the ability of the adjoint method to recover parameters. The success of the twin experiment is judged by the agreement between the optimal values and the base values of the parameters (Spitz *et al.* 1998). A previous study shows that monthly averaged chlorophyll-*a* data are not sufficient to recover parameter values (Xu *et al.* 2006) for the NPZD model in the study area. For convenience, 5-day averages of chlorophyll-*a* data with the same horizontal resolution as that in the NPZD model are assimilated into the model in the present study.

As reported by Matear (1995), the inherent nonlinearities associated with most ecosystem models, suggest that many parameters are correlated with each other. If correlated parameters are considered as control variables in an adjoint assimilation experiment, the system may become under-determined, and large uncertainties in the optimal parameter values may appear (Friedrichs 2001a). Therefore, it is crucial to determine which parameters should be chosen as the control variables.

Twin experiment 1

The first issue is to investigate if all the model parameters can be recovered. Thus all the model parameters are considered as the control variables first. The model parameters have different orders of magnitude (see Table 1). To avoid precision problems with the data assimilation, these values were scaled so that the control variables have more uniform magnitudes (Lawson *et al.* 1996). Hence the parameter X_i was replaced by $X'_i X_i$ in the model ($i = 1, \dots, 14$). A recovery of $X'_i = 1$ corresponds to an optimal value for the corresponding model parameter.

Garcia-Gorriz *et al.* (2003) indicated that different dynamics/periods produce different sets of the optimal values of the control variables. To take seasonal variations of the parameters in the assimilation of real data into account, the optimal values of the control variables are estimated every 5 days with the same set of their first-guess values. We independently perturb all the parameters over random error distributions from 20% to 50% of base values (in Table 1), and employ these as first-guess values for the assimilation experiment.

Control variable choosing

In this section, the sensitivity analysis is conducted to assess the correlation between model parameters and then to choose uncorrelated parameters as control variables for the assimilation experiment.

The sensitivity $S_{C,K}$ of a certain state variable C to a given model parameter K is defined as the fractional change in C as a result of a fractional increase in the value of K (Friedrichs 2001a; Garcia-Gorriz *et al.* 2003).

$$S_{C,K} = \left(\frac{C_K - C_{K+\%}}{C_{K+\%}} \right) / \left(\frac{K - K_{+\%}}{K_{+\%}} \right), \quad (2)$$

where K is the value of the parameter; and $K_{+\%}$ denotes the value of the parameter increased by a fractional of its reference value K . The terms C_K and $C_{K+\%}$ are the concentration values of nutrient (N), phytoplankton (P), zooplankton (Z) or detritus (D), computed with the parameter value K and $K_{+\%}$ respectively.

Referring to Garcia-Gorriz *et al.* (2003), we carried out lots of numerical experiments with different percentages of increase (10%, 20% and 25%), with different first-guess values of model parameters and at different spatial locations in the study area. According to the sensitivity analysis, five unrelated rate parameters can be chosen as control variables, namely V_m , G_m , d_p , d_z and e (see Discussion section for control variable choosing).

Twin experiment 2

A similar twin experiment was carried out as that in twin experiment 1, except that the control variables here are the five parameters V_m , G_m , d_p , d_z and e selected by the sensitivity analysis. This twin experiment was conducted to investigate if the adjoint method is capable of recovering the control variables. All five control variables are also perturbed over random error distributions from 20% to 50% of base values in Table 1.

Assimilation of remote-sensed chlorophyll-*a* data

As a preliminary study on the ability of the adjoint method to improve model outputs via assimilating real data, the true remote sensed sea surface chlorophyll-*a* concentration data are assimilated into the NPZD model.

Similar to numerical twin experiments, the assimilated data in this section are 5-day averaged sea surface chlorophyll-*a* concentration data with the same horizontal resolution as that in the NPZD model. Monthly averaged blended chlorophyll-*a* data provided by the National Satellite Ocean Application Service (NSOAS) of China are interpolated to generate 5-day averaged data. The monthly averaged data are fused by using the Kriging method from three resources including the Sea-viewing Wide Field-of-view Sensor (SeaWiFS) onboard SeaStar, Moderate Resolution Imaging Spectroradiometer (MODIS) onboard EOS-AM and Chinese Ocean Colour and Temperature Scanner (COCTS) onboard HY-1. Lin (Lin, M. S., pers. comm.) pointed out that the precision of blended data is greatly improved compared with those retrieved from the single satellite sensor.

The chlorophyll-*a* data are converted from the unit of mg chlorophyll-*a* m^{-3} to the model unit of mmol N m^{-3} , using

an assumed constant chlorophyll-*a* to nitrogen ratio of 2.5 mg chlorophyll-*a* (mmol N)⁻¹, which is calculated using a constant carbon to nitrogen (C:N; mmol C:mmol N) ratio for phytoplankton of 6.6 (Friedrichs 2001b) and a carbon to chlorophyll-*a* (mg C:mg chlorophyll-*a*) ratio of 32 (Zhao 2002).

Base values in the twin experiments are listed in Table 1 are used for the first-guess values of the five control variables. Similar to numerical twin experiments, a set of optimal values of control variables is recovered every 5 days in 2003.

Results

Analysis of the physical model simulation

To show the reliability of the physical model, the simulated sea surface temperature (SST) was compared with monthly observations from Reynolds/NCEP and MODIS/Aqua (<http://poet.jpl.nasa.gov/>) in August 2003 (Fig. 3). It is shown in Fig. 3 that in the BS, the water was warmer in the Bohai Bay where the depth is relatively shallower. It then became colder in the central area and a cold water core occurred in the south coast of Liaodong Peninsula. In the NYS, the temperature was steadily changed. The result indicates that the model can capture the main feature of SST distribution in summer. Figure 4 is the sea surface current in winter and summer respectively. Since the BS and NYS is an area with strong tidal currents, the monthly current field is relatively weak. Figure 4 shows that the surface current in the study area has the characteristic of wind current. That is, in winter, the surface water mainly flowed to the south when the northerly wind was prevailing, whereas in summer it flowed to the north.

Analysis of numerical twin experiment 1

Results for the cases in which the first-guess values of all the parameters are increased by 20% (Case 1) and 50% (Case 2) for the first 5 days in the year, respectively, are analysed. Table 2 lists the scaled base values and the optimal values of the parameters for Case 1 and Case 2 respectively. The number in brackets denotes the percentage error between the optimal value and base value computed as

$$100 \times \frac{|\text{optimal value} - \text{base value}|}{\text{base value}}$$

Table 2 shows that all of the parameters can be appropriately recovered when the base values of the parameters are increased by 20%, whereas they cannot be estimated when the base values are increased by more than 50%. The other 5-day experiments in the year lead to similar results. This is consistent with the investigations of Friedrichs (2001a) who pointed out that if all of the correlated model parameters were to be estimated by the adjoint method, large uncertainties in the optimal values may appear. Thus control variables have to be chosen first before the application of the adjoint method to the coupled model.

Analysis of control variable choosing

Results of sensitivity analysis indicate that the value of the sensitivity of the state variable to a given model parameter is hardly affected by the fractional increase of the parameter value,

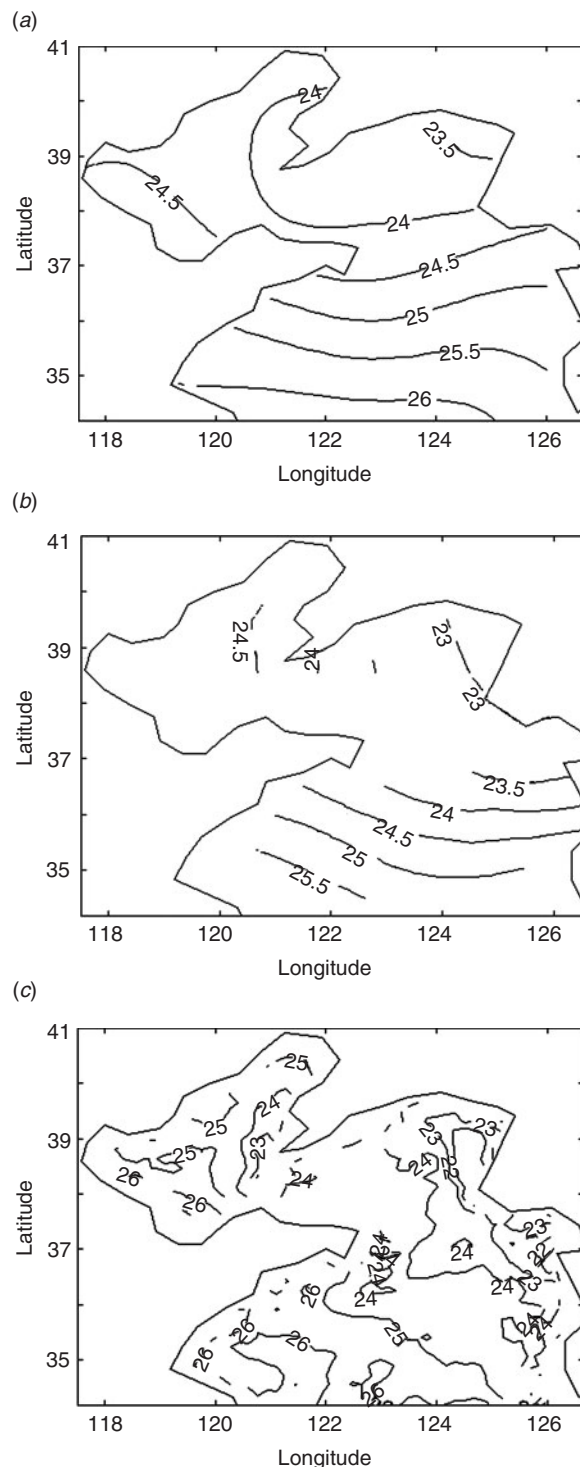


Fig. 3. Comparison of model simulated sea surface temperature (SST) (a) with monthly observations from Reynolds/National Centers for Environmental Prediction (NCEP) (b) and Moderate Resolution Imaging Spectroradiometer (MODIS)/Aqua (c) in August 2003.

the spatial location, or the first-guess value of the parameter. Therefore, we list only the values of the sensitivity of the yearly-averaged state variables for the study area, with the fractional increase of 25% of each model parameter value.

The values of the sensitivity for nutrient (N), phytoplankton (P), zooplankton (Z) and detritus (D) (i.e. S_N , S_P , S_Z and S_D) are listed in Table 3. The default symbol ‘–’ denotes the absolute value of the sensitivity to the parameter is less than 0.05. Similarities in the absolute magnitudes of S_N , S_P , S_Z and S_D (Table 3) for G_m (maximum grazing rate of P by Z), λ (Ivlev grazing constant) and BQ_{10} (temperature adjustment coefficient for growth of Z) indicate that these three parameters are highly correlated. Similarly, the parameters γ (assimilation coefficient of Z) and d_Z (mortality rate of Z) are also correlated. The two sets of correlated parameters are both related with the zooplankton. For the phytoplankton-related parameters, the sensitivities of the four state variables to V_m (maximum uptake rate of nutrient by P) are approximately three times that of k_s (half-saturation constant of nitrogen by P) and AQ_{10} (temperature adjustment coefficient for growth of P), and approximately seven times and 10 times that of w_P and k_{ext} , respectively, indicating that these five parameters are correlated. In addition, parameters d_P (mortality rate of P) and e (regeneration rate of D) are relatively independent of the other parameters, and w_D and θ (excreting rate of Z) display very small sensitivities.

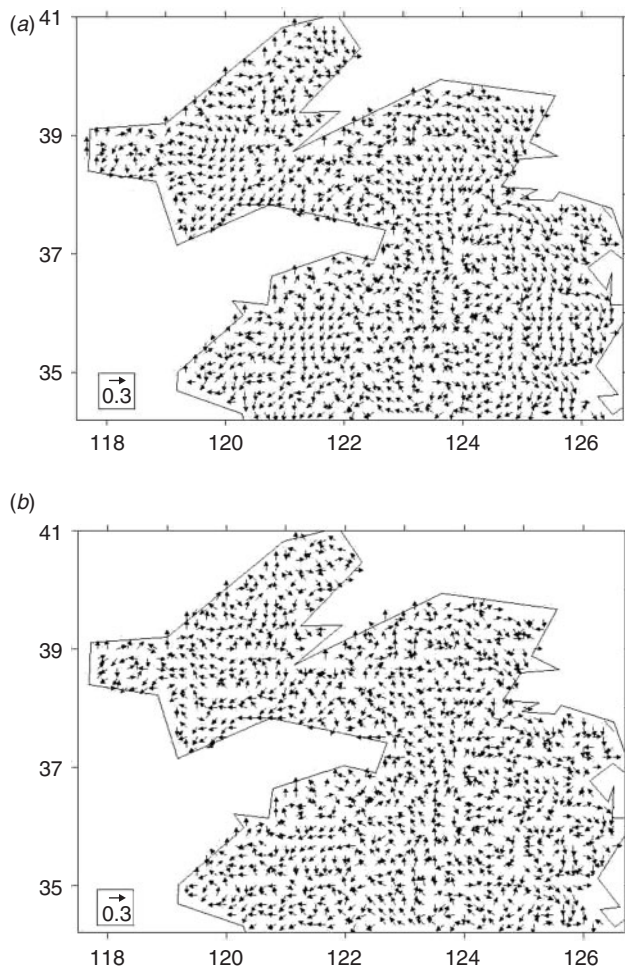


Fig. 4. Model simulated sea surface current in the Bohai Sea (BS) and Northern Yellow Sea (NYS) in February (a) and August (b) 2003.

Analysis of numerical twin experiment 2

For twin experiment 2, results for the cases in which the first-guess values of all of the five parameters are increased by 20% (Case 1) and 50% (Case 2) for the first 5-day experiment are also analysed. The scaled base values and optimal values of the control variables for Case 1 and Case 2, respectively, are listed in Table 4. The number in brackets denotes the percent error

Table 2. Scaled parameter values before and after assimilation in twin experiment 1

Parameter	Base value	Optimal value for Case 1 (error)	Optimal value for Case 2 (error)
V_m'	1.0	1.004 (0.4%)	1.183 (18.3%)
k_s'	1.0	0.997 (0.3%)	1.194 (19.4%)
k_{ext}'	1.0	1.001 (0.1%)	1.217 (21.7%)
w_P'	1.0	1.002 (0.2%)	1.240 (24.0%)
d_P'	1.0	1.003 (0.3%)	1.161 (16.1%)
AQ_{10}'	1.0	1.003 (0.3%)	1.089 (8.9%)
λ'	1.0	0.989 (1.1%)	1.254 (25.4%)
G_m'	1.0	1.038 (3.8%)	1.246 (24.6%)
γ'	1.0	1.017 (1.7%)	1.163 (16.3%)
θ'	1.0	1.003 (0.3%)	1.240 (24.0%)
d_Z'	1.0	1.003 (0.3%)	1.282 (28.2%)
BQ_{10}'	1.0	1.003 (0.3%)	1.055 (5.5%)
e'	1.0	1.008 (0.8%)	1.256 (25.6%)
w_D'	1.0	1.003 (0.3%)	1.134 (13.4%)

Table 3. The sensitivity of state variables to model parameters

Parameter	S_N	S_P	S_Z	S_D
V_m	−0.55	0.90	0.22	–
k_s	0.17	−0.28	−0.07	–
k_{ext}	0.05	−0.09	−0.02	–
AQ_{10}	−0.19	0.29	0.08	–
w_P	0.07	−0.13	−0.03	–
G_m	0.06	−0.25	0.21	–
λ	0.05	−0.21	0.18	–
BQ_{10}	0.05	−0.20	0.19	–
γ	–	−0.07	0.33	–
d_Z	–	0.08	−0.40	–
d_P	0.05	−0.20	–	–
e	0.11	0.10	–	−0.10
w_D	–	–	–	–
θ	–	–	–	–

Table 4. Scaled parameter values before and after assimilation in twin experiment 2

Parameter	Base value	Optimal value for Case 1 (error)	Optimal value for Case 2 (error)
V_m'	1.0	1.005 (0.5%)	1.004 (0.4%)
d_P'	1.0	1.0 (0.0%)	1.0002 (0.02%)
G_m'	1.0	1.0046 (0.46%)	1.004 (0.4%)
d_Z'	1.0	1.0 (0.0%)	1.004 (0.4%)
e'	1.0	1.0034 (0.34%)	1.004 (0.4%)

between the optimal value and base value. Table 4 shows that all five control variables are perfectly recovered regardless of the percentage that the base values of the control variables are disturbed.

Figure 5 shows the value of the ratio of the cost function to its initial value as a function of the number of iterations for Case 2. The cost function is reduced from $O(10^0)$ to $O(10^{-4})$ in roughly

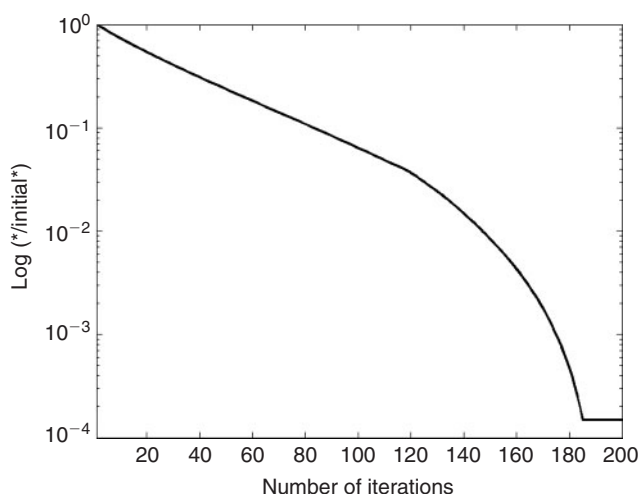


Fig. 5. Value of the ratio of the cost function to its initial value for Case 2 in twin experiment 2 (y-axis denotes logarithmic ordinate).

200 iterations. Figure 6 shows the variations of the scaled five control variables, expressed by the ratio between optimised value and base value, as a function of the number of iterations in the twin experiment for Case 2. It is shown that the regeneration rate of detritus e is first recovered, phytoplankton mortality rate of phytoplankton d_p , maximum grazing rate of phytoplankton by zooplankton G_m and mortality rate of zooplankton d_z follow, and maximum uptake rate of nutrient by phytoplankton V_m is the last recovered parameter. The values of all of the control variables are perfectly recovered.

Analysis of assimilation of remote sensed chlorophyll-a data

Figure 7 shows the averaged ratio of the cost function to its initial value during the year 2003 as a function of the number of iterations. Abscissa denotes the number of iterations. The value of the cost of function is significantly reduced as a result of changes in the values of control variables, suggesting that the misfit between the simulations and observations becomes much smaller, and the adjoint method is effective.

A comparison between the observations, the model outputs before data assimilation and those after assimilation is shown in Fig. 8. The solid line denotes the annual cycle of remote sensed chlorophyll-*a* concentrations, the dashed line denotes model outputs before assimilation, and the dash-dotted line denotes those after assimilation. Abscissa is time in months. It is clearly shown that there is a perfect agreement between model outputs after

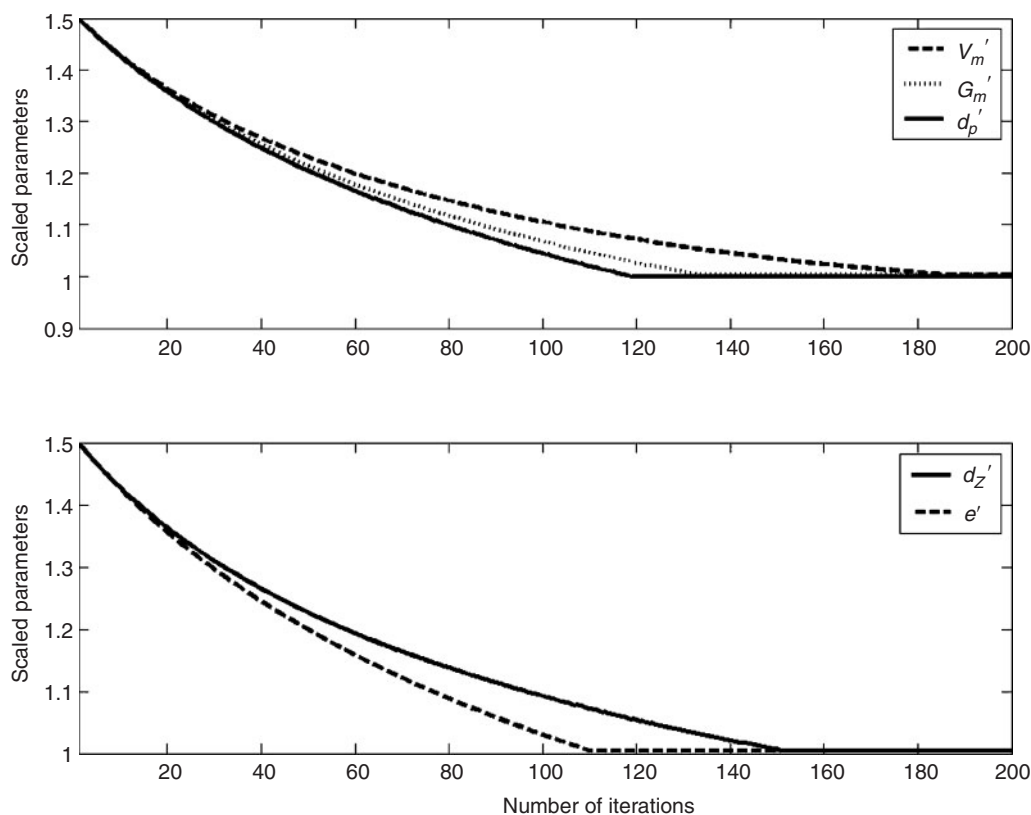


Fig. 6. Variations of scaled five control variables for Case 2 in twin experiment 2.

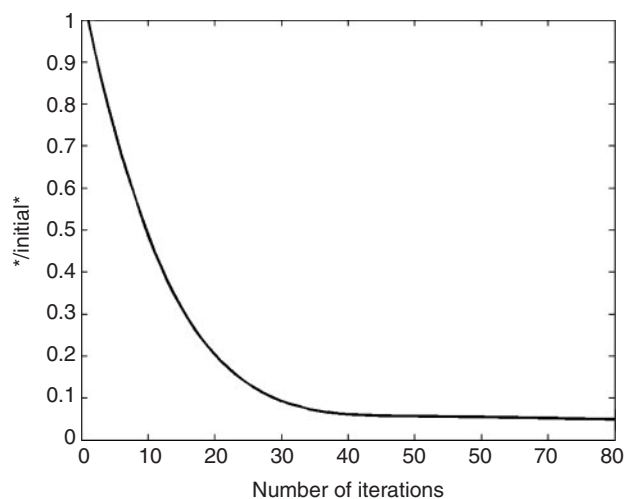


Fig. 7. Average ratio of the cost function to its initial value.

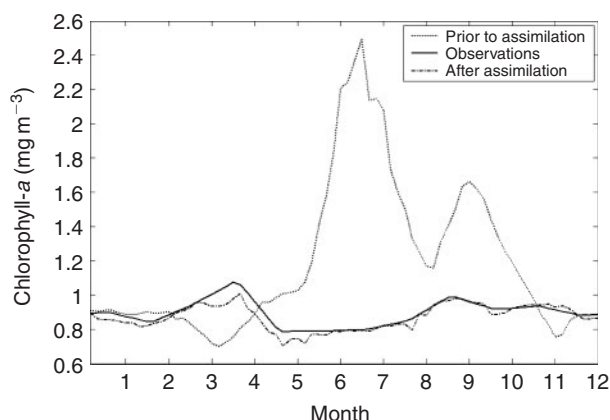


Fig. 8. Annual cycle of surface chlorophyll-*a* concentrations in 2003.

assimilation and the observations, and the model has effectively mimicked the phytoplankton blooms in spring and autumn in 2003.

To further show the assimilation results, sea surface concentrations of monthly-averaged chlorophyll-*a* in April and September, when the phytoplankton blooms occur, are plotted in Fig. 9. The upper three figures are the monthly-averaged chlorophyll-*a* concentrations of the observations, model outputs before assimilation and those after assimilation in April for the studied domain respectively. The lower three figures are corresponding to sea surface concentrations of monthly-averaged chlorophyll-*a* in September for the three cases respectively.

A more significant result of the assimilation is shown by a comparison between the model outputs of monthly mean surface nutrient (nitrate in the present study) concentrations and *in situ* monthly-mean observations in the study area provided by National Marine Data and Information Service (NMDIS) of China, which is plotted in Fig. 10. The solid line denotes the annual cycle of nutrient concentrations of observations, the dash-dotted line denotes model outputs before assimilation, and the dotted line denotes those after assimilation. It is clear

that there is a better agreement between model outputs after assimilation and the observations.

Discussion

Numerical twin experiments and control variable choosing

Numerical twin experiment 1 shows that not all of the model parameters can be selected as control variables in the application of the adjoint method. According to the results of the sensitivity analysis (Table 3), we can choose any one from each set of correlated parameters (Friedrichs 2001a). The chosen three parameters from the three sets of correlated parameters, together with the two independent parameters, may consist of a set of control variables in the NPZD model.

At the same time, the gradients of the cost function with respect to the parameters were also calculated. The result shows that the impacts of the five rate parameters V_m , G_m , d_P , d_Z and e on the cost function are much larger than those of the other parameters, suggesting that the five parameters are more important compared with the others. They happen to constitute a set of control variables from the sensitivity analysis. Li and Wunsch (2004) pointed out the larger the impact of the model parameter on the cost function; the larger the parameter is confined to the observations, indicating that the parameter can be recovered more accurately. Thus the five rate parameters V_m , G_m , d_P , d_Z and e were chosen as control variables.

With these parameters as control variables, the adjoint method shows the capability of perfectly recovering their values. Both Fig. 5 and Fig. 6 indicate that quantity of data available is enough for the cost function and control variables to converge and that the adjoint model is also correctly constructed, because the optimal values of control variables obtained from this assimilation are equal to their base values. Similar results are obtained for the other 5-day experiments. This verifies the feasibility of the adjoint method to assimilate remote sensed chlorophyll-*a* data.

Assimilation of remote sensed chlorophyll-*a* data

To evaluate qualitatively the performance of the data assimilation of remote sensed chlorophyll-*a* data, three quantities are computed, i.e. the misfit between model outputs and observations, the RMS error and the correlation coefficient. The misfit ξ is defined as (Garcia-Gorriz et al. 2003).

$$\xi = 100 \left| 1 - \frac{C_M}{C_0} \right|, \quad (3)$$

where C_0 and C_M are the observations and model outputs, respectively, averaged over time and space.

The misfit and RMS error are a measure of the difference between model outputs and observations, whereas the correlation coefficient gives a measure of the phase shift. It is important to consider all quantities to evaluate the success of the data assimilation. The misfit between model outputs before assimilation and the observations in Fig. 8 is 32.83%, the RMS error is 0.567 mg m⁻³, and the correlation coefficient is 0.47; whereas the misfit and RMS error between model outputs after assimilation and the observations are significantly reduced to 2.57% and 0.038 mg m⁻³, respectively, and the correlation coefficient

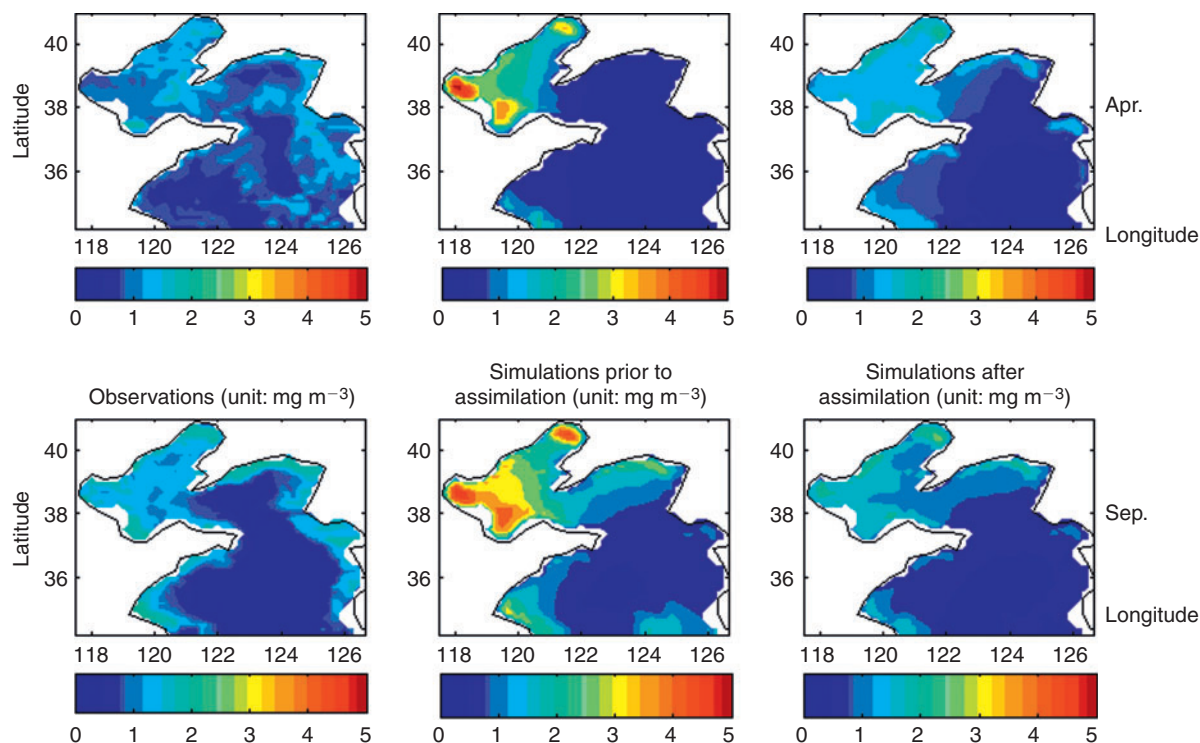


Fig. 9. Sea surface chlorophyll-*a* concentrations in April and September 2003.

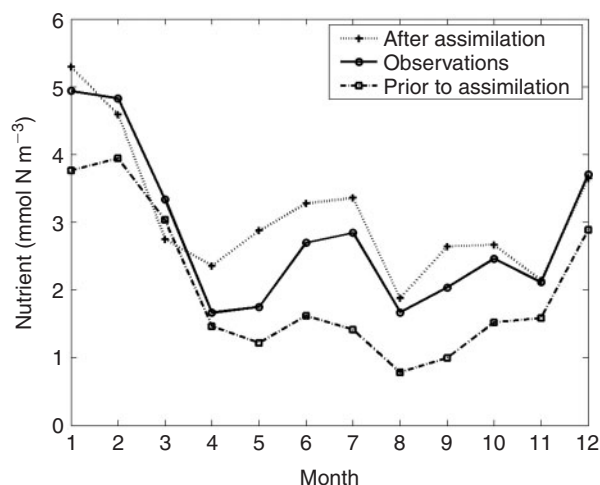


Fig. 10. Annual cycle of sea surface nutrient concentrations in 2003.

increases to 0.91. This indicates that the model outputs are remarkably improved when the adjoint method is applied to the NPZD model.

In Fig. 9, for April, the misfit and RMS error between model outputs before assimilation and the observations are 23.67% and 0.922 mg m^{-3} , respectively, and those between model outputs after assimilation and the observations are 8.21% and 0.439 mg m^{-3} respectively. For September, the two misfits are 49.24% and 0.37%, respectively, and the two RMS errors are 0.964 mg m^{-3} and 0.450 mg m^{-3} respectively. The results further show that the phytoplankton blooms for the BS and NYS

are well mimicked by the NPZD model with the application of the adjoint method, especially for September.

In Fig. 10, whereas the correlation coefficient is little changed, the misfit and RMS error are reduced from 28.84% and $0.88 \text{ mmol N m}^{-3}$ before assimilation to 10.10% and $0.52 \text{ mmol N m}^{-3}$ after assimilation, respectively, which also shows the capability of the adjoint method in improving the simulation.

Conclusions

A 3-D NPZD model coupled with the POM model is developed for the Bohai and the Northern Yellow Sea. Control variables are determined by the sensitivity analysis. Based on the success of the numerical twin experiments, remote sensed chlorophyll-*a* data are assimilated into the NPZD model, leading to a better agreement between model outputs and the observations. Thus the model can describe the phytoplankton bloom conditions and the annual cycle of phytoplankton and nutrient fairly well.

Numerical twin experiments are optimal since they are not confronted with measurement errors and inconsistencies between the observations and the model. The addition of error associated with the observations, however, can hinder the rate of parameter recovery (Lawson *et al.* 1995). In the further study, the uncertainty induced by several sources of error should be evaluated.

In the present study, only preliminary results of adjoint assimilation of real data at the sea surface layer are presented, the vertical profiles are to be discussed in a further study. Furthermore, the chlorophyll-*a* to nitrogen ratio in the phytoplankton is

assumed constant with time. As reported by Spitz *et al.* (1998), during periods of low light, such as winter, the ratio is high, and during periods of high light, such as summer, it is low. This seasonal variation in the chlorophyll-*a* to nitrogen ratio was used by Hurtt and Armstrong (1996) to improve their ecosystem model. Thus in future analysis, the seasonal variation in the chlorophyll-*a* to nitrogen ratio is to be considered.

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Appendix A

NPZD model

In the nitrogen-based NPZD (nitrate, phytoplankton, zooplankton and detritus) pelagic model, the evolution of each state variable is determined by an advective-diffusive equation of the physical model as follows.

$$\begin{aligned} \frac{\partial P}{\partial t} + u \frac{\partial P}{\partial x} + v \frac{\partial P}{\partial y} + w \frac{\partial P}{\partial z} + w_p \frac{\partial P}{\partial z} &= \frac{\partial}{\partial x} \left(A_p \frac{\partial P}{\partial x} \right) \\ &+ \frac{\partial}{\partial y} \left(A_p \frac{\partial P}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_p \frac{\partial P}{\partial z} \right) + \frac{V_m N}{k_s + N} \cdot e^{-k_{ext} z} \\ &\cdot A Q_{10}^{(T-10)/10} \cdot P - G_m \cdot B Q_{10}^{(T-10)/10} \\ &\cdot (1 - e^{-\lambda P}) \cdot Z - d_P P \end{aligned} \quad (A-1)$$

$$\begin{aligned} \frac{\partial Z}{\partial t} + u \frac{\partial Z}{\partial x} + v \frac{\partial Z}{\partial y} + w \frac{\partial Z}{\partial z} &= \frac{\partial}{\partial x} \left(A_p \frac{\partial Z}{\partial x} \right) \\ &+ \frac{\partial}{\partial y} \left(A_p \frac{\partial Z}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_p \frac{\partial Z}{\partial z} \right) + \gamma G_m \\ &\cdot B Q_{10}^{(T-10)/10} \cdot (1 - e^{-\lambda P}) \cdot Z - d_Z Z \end{aligned} \quad (A-2)$$

$$\begin{aligned} \frac{\partial N}{\partial t} + u \frac{\partial N}{\partial x} + v \frac{\partial N}{\partial y} + w \frac{\partial N}{\partial z} &= \frac{\partial}{\partial x} \left(A_p \frac{\partial N}{\partial x} \right) \\ &+ \frac{\partial}{\partial y} \left(A_p \frac{\partial N}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_p \frac{\partial N}{\partial z} \right) - \frac{V_m N}{k_s + N} \cdot e^{-k_{ext} z} \\ &\cdot A Q_{10}^{(T-10)/10} P + \theta \cdot G_m \cdot B Q_{10}^{(T-10)/10} \\ &\cdot (1 - e^{-\lambda P}) \cdot Z + e D \end{aligned} \quad (A-3)$$

$$\begin{aligned} \frac{\partial D}{\partial t} + u \frac{\partial D}{\partial x} + v \frac{\partial D}{\partial y} + w \frac{\partial D}{\partial z} + w_D \frac{\partial D}{\partial z} &= \frac{\partial}{\partial x} \left(A_p \frac{\partial D}{\partial x} \right) \\ &+ \frac{\partial}{\partial y} \left(A_p \frac{\partial D}{\partial y} \right) + \frac{\partial}{\partial z} \left(K_p \frac{\partial D}{\partial z} \right) + (1 - \gamma - \theta) G_m \\ &\cdot B Q_{10}^{(T-10)/10} \cdot (1 - e^{-\lambda P}) \cdot Z + d_P P + d_Z Z - e D \end{aligned} \quad (A-4)$$

where P , Z , N and D are phytoplankton, zooplankton, nutrient and detritus, respectively; u , v and w are the 3-D water velocity determined by the physical model; T is the temperature of water; A_p and K_p are the horizontal and vertical eddy diffusivities respectively. The model parameters are described in Table 1.

Appendix B

The adjoint equations of the NPZD model

$$\begin{aligned} -\frac{\partial P^*}{\partial t} - \frac{\partial}{\partial z} \left(K_p \frac{\partial P^*}{\partial z} \right) &= \frac{\partial}{\partial x} (u P^*) + \frac{\partial}{\partial y} (v P^*) + \frac{\partial}{\partial z} (w P^*) \\ &+ w_p \frac{\partial P^*}{\partial z} + \frac{\partial}{\partial x} \left(A_p \frac{\partial P^*}{\partial x} \right) + \frac{\partial}{\partial y} \left(A_p \frac{\partial P^*}{\partial y} \right) + \frac{V_m N}{k_s + N} \\ &\cdot e^{-k_{ext} z} \cdot A Q_{10}^{(T-10)/10} (P^* - N^*) - d_P \cdot (P^* - D^*) \\ &- \lambda \cdot e^{-\lambda P} \cdot G_m B Q_{10}^{(T-10)/10} \cdot Z [P^* - \gamma \cdot Z^* - \theta \cdot N^* \\ &- (1 - \gamma - \theta) \cdot D^*] - (P - \bar{P}) \end{aligned} \quad (B-1)$$

$$\begin{aligned} -\frac{\partial Z^*}{\partial t} - \frac{\partial}{\partial z} \left(K_p \frac{\partial Z^*}{\partial z} \right) &= \frac{\partial}{\partial x} (u Z^*) + \frac{\partial}{\partial y} (v Z^*) + \frac{\partial}{\partial z} (w Z^*) \\ &+ \frac{\partial}{\partial x} \left(A_p \frac{\partial Z^*}{\partial x} \right) + \frac{\partial}{\partial y} \left(A_p \frac{\partial Z^*}{\partial y} \right) - G_m \cdot B Q_{10}^{(T-10)/10} \\ &\times (1 - e^{-\lambda P}) [P^* - \gamma \cdot Z^* - \theta \cdot N^* - (1 - \gamma - \theta) \cdot D^*] \\ &- d_Z \cdot (Z^* - D^*) - (Z - \bar{Z}) \end{aligned} \quad (B-2)$$

$$\begin{aligned} -\frac{\partial N^*}{\partial t} - \frac{\partial}{\partial z} \left(K_p \frac{\partial N^*}{\partial z} \right) &= \frac{\partial}{\partial x} (u N^*) + \frac{\partial}{\partial y} (v N^*) + \frac{\partial}{\partial z} (w N^*) \\ &+ \frac{\partial}{\partial x} \left(A_p \frac{\partial N^*}{\partial x} \right) + \frac{\partial}{\partial y} \left(A_p \frac{\partial N^*}{\partial y} \right) + \frac{V_m k_s}{(k_s + N)^2} \\ &\cdot e^{-k_{ext} z} \cdot A Q_{10}^{(T-10)/10} \cdot P \cdot (P^* - N^*) - (N - \bar{N}) \end{aligned} \quad (B-3)$$

$$\begin{aligned} -\frac{\partial D^*}{\partial t} - \frac{\partial}{\partial z} \left(K_p \frac{\partial D^*}{\partial z} \right) &= \frac{\partial}{\partial x} (u D^*) + \frac{\partial}{\partial y} (v D^*) + \frac{\partial}{\partial z} (w D^*) \\ &+ w_D \frac{\partial D^*}{\partial z} + \frac{\partial}{\partial x} \left(A_p \frac{\partial D^*}{\partial x} \right) + \frac{\partial}{\partial y} \left(A_p \frac{\partial D^*}{\partial y} \right) \\ &+ e \cdot (N^* - D^*) - (D - \bar{D}) \end{aligned} \quad (B-4)$$

where N^* , P^* , Z^* and D^* are the corresponding Lagrange adjoint operators of N , P , Z and D , respectively; $\bar{N}$, $\bar{P}$, $\bar{Z}$ and $\bar{D}$ are the corresponding available observations to be assimilated.